EmoTherm SecureChain

A Privacy-Preserving and Tamper-Evident Framework
for Thermal Facial-Expression Research

Teacher Presentation and Viva Briefing

Problem, workflow, technology stack, algorithms, security, verification, results, limitations, and demonstration sequence

Authors

Darshan A, Dhruva N, Sohan H S, Suhas U M

Department of Computer Science and Business Systems

Malnad College of Engineering, Hassan, Karnataka, India

Under the Guidance of Dr. C M Naveen Kumar

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1. Explain the project in two minutes

Use this answer when the teacher first asks: What is your project?

Suggested opening

Our project, EmoTherm SecureChain, is a full-stack research platform that accepts a thermal face image from an upload or camera, preprocesses it, estimates one of five facial-expression classes, encrypts both the image and result, and stores a cryptographic commitment in a private hash-linked blockchain. Every result can later be verified and exported as a digitally signed PDF and JSON certificate. The main contribution is the complete trustworthy pipeline: acquisition, classification, privacy, provenance, access control, audit, and independent verification.

The five supported labels are **disgust, happy, neutral, sad, and surprise**. The active classifier uses Histogram of Oriented Gradients (HOG) with multinomial logistic regression. The current fixed image-level holdout accuracy is **83.6 percent** on 165 images. This number does not establish accuracy for completely new people because subject identities were not controlled across the split.

The system must be described as a **thermal facial-expression research platform**, not a medical emotion detector. It estimates similarity to trained expression labels; it cannot prove a person's internal emotional state.

What makes the project broader than classification

- Thermal upload and three camera paths: browser UVC, OpenCV bridge, and serial/vendor adapter interface.
- Secure user accounts, roles, revocable sessions, and record ownership.
- Per-record AES encryption with RSA key wrapping.
- Tamper-evident SHA-256 private blockchain and audit log.
- Signed verification certificates with QR code and public verification page.
- Measured model metrics, history, administration, and reproducible tests.

2. Problem statement, motivation, and objectives

Problem statement

Thermal facial images are sensitive research evidence. A conventional classifier may return a label, but it usually does not protect the image, establish who created the result, record which model was used, or detect later alteration. The project addresses recognition and evidence integrity in one system.

Why thermal imaging?

- It captures infrared intensity or temperature-related facial patterns rather than ordinary visible color.
- It can operate in low visible-light conditions when the chosen sensor supports the required capture.
- It introduces privacy, calibration, and domain-shift challenges that make provenance especially important.

Project objectives

Objective	Implemented result
Acquire	Upload or capture thermal frames through browser or Python bridge.
Recognize	Preprocess and classify into five trained expression classes.
Protect	Encrypt raw evidence and predictions before database storage.
Prove	Commit ciphertext and provenance to a hash-linked chain.
Control	Enforce authentication, ownership, roles, rate limits, and audit logs.
Verify	Recompute cryptographic commitments and validate the whole chain.
Export	Produce signed PDF/JSON certificates and QR-based verification.

3. System architecture

The application is layered so that each concern can be explained and tested independently.

1. Input	2. Validate	3. Preprocess	4. Classify
Upload / UVC / bridge	MIME, encoding, dimensions, size	Blur, normalize, CLAHE, resize	HOG plus logistic regression
5. Encrypt	6. Commit	7. Persist	8. Verify
AES-GCM plus RSA-OAEP	SHA-256 plus private chain	SQLite/PostgreSQL transaction	GCM, hash, chain, signature

Layer responsibilities

Layer	Technology	Responsibility
Presentation	React, React Router, Recharts, Framer Motion	Pages, forms, charts, camera UI, downloads.
API	FastAPI, Uvicorn, Pydantic	Validation, authentication, inference orchestration, REST and WebSocket.
Data	SQLAlchemy, SQLite / PostgreSQL	Users, sessions, encrypted uploads, predictions, blocks, audit records.
Machine learning	OpenCV, NumPy, scikit-learn, PyTorch baseline	Preprocessing, HOG features, logistic classifier, metrics; original CNN retained for comparison.
Cryptography	cryptography library	AES-GCM, RSA-OAEP, RSA-PSS, SHA-256.
Integrity	Python private chain	Canonical hashing, previous-hash links, proof-of-work demonstration.
Documents	ReportLab	Downloadable QR-enabled PDF certificate.

One-line architecture answer

React provides the interface, FastAPI controls the workflow, the machine-learning module estimates the class, cryptography protects the evidence, SQLAlchemy persists it, and the private chain plus signed certificate makes later verification possible.

4. End-to-end workflow: what happens at each step

Step	Operation	Technology / algorithm	Output
1	User signs in	bcrypt password hash; JWT access token; rotating HttpOnly refresh token	Authorized session
2	Image enters	Upload, browser canvas, OpenCV, serial, or vendor adapter	Raw image bytes
3	Input validation	MIME/encoding match, maximum 10 MB, dimension and pixel limits	Accepted thermal image
4	Preprocessing	Grayscale, 3x3 Gaussian blur, min-max normalization, CLAHE, 96x96 resize	Normalized model input
5	Feature extraction	HOG: 8x8 cells, 16x16 blocks, 9 gradient orientations	Numerical feature vector
6	Classification	Multinomial logistic regression, C=10, softmax probability	Five probabilities and winning label
7	Evidence encryption	Fresh AES-256 key; independent GCM nonces; scan UUID as AAD	Authenticated ciphertext
8	Key protection	RSA-3072 OAEP-SHA256 wraps the AES key	Wrapped per-record key
9	Commitment	SHA-256 of encrypted image, encrypted prediction, and canonical metadata	Record commitment
10	Block append	Previous hash plus nonce; demonstration prefix 00	Tamper-evident block
11	Atomic storage	SQLAlchemy transaction	Upload, prediction, block, audit event
12	Verification/export	GCM authentication, recomputed hashes, chain validation, RSA-PSS signature	Verified result and PDF/JSON certificate

5. Image preprocessing algorithms

A. Grayscale conversion

Thermal inputs are represented as a single intensity channel. Palette-colored exports are converted to grayscale, so the active model learns intensity structure rather than RGB color. For a truly radiometric camera, raw temperature values should be preserved and calibrated rather than reduced from a display palette.

B. Gaussian smoothing

A 3 by 3 Gaussian filter suppresses high-frequency sensor noise. Conceptually, every output pixel is a weighted average of its neighbors, with larger weights near the center. This reduces isolated noise before gradient calculation.

C. Min-max normalization

Pixel intensities are mapped to the range 0 to 255 using: **I' = 255 (I - Imin) / (Imax - Imin)**. This improves consistency across stored images, but it also removes absolute temperature information.

D. CLAHE

Contrast Limited Adaptive Histogram Equalization divides the image into local tiles, improves local contrast, and clips excessive histogram amplification. It helps reveal facial edges without allowing noise to dominate.

E. Resize

Every image is resized to 96 by 96 pixels. A fixed size produces a consistent HOG feature length and predictable inference time.

What to say if asked why preprocessing matters

A model must receive data in the same representation used during training. Preprocessing reduces noise and shape differences, improves local thermal contrast, and fixes the input dimensions. A mismatch between training and live-camera preprocessing causes domain shift and wrong predictions.

6. Active recognition algorithm: HOG plus logistic regression

Histogram of Oriented Gradients (HOG)

HOG describes local shape. It calculates horizontal and vertical intensity gradients, converts them into magnitude and orientation, creates orientation histograms inside small cells, and normalizes neighboring cells in blocks. The implementation uses **8x8 pixel cells, 16x16 pixel blocks, and 9 orientation bins** on the 96x96 preprocessed image.

HOG sequence: **gradients -> magnitude/orientation -> cell histograms -> block normalization -> concatenated feature vector**.

Multinomial logistic regression

For every class k, the classifier calculates a linear score **zk = wk.x + bk**. Softmax converts all scores to probabilities: **P(k|x) = exp(zk) / sum exp(zj)**. The class with the largest probability is returned. The regularization parameter is **C = 10**, and balanced class weights reduce the effect of unequal class counts.

Why this replaced the original CNN

The original small PyTorch CNN underfit the supplied dataset and achieved 35.8 percent held-out accuracy. HOG plus logistic regression achieved 83.6 percent on the same style of fixed image-level holdout. The original CNN remains available as a baseline for comparison; the application uses **thermal-hog-logreg-v2**.

What confidence means

Confidence is the classifier probability for the selected training label. It is not biological certainty. A high value can still be wrong on a new person or different camera, especially when the new input differs from the training distribution.

7. Dataset, training, and measured evaluation

Class	Usable images
Happy	215
Sad	201
Neutral	198
Disgust	142
Surprise	69
Total	825

Dataset cleaning

- 838 thermal files were discovered.

- Exact file duplicates and unreadable images were excluded.

- 825 usable images remained.

- The source dataset was not modified.

Evaluation configuration

The active evaluation uses a fixed stratified 80/20 image split with seed 42: 660 training images and 165 held-out test images. The deployment classifier is subsequently refit on all 825 usable images.

Metric	Result
Test accuracy	83.6% (138 of 165)
Macro precision	0.795
Macro recall	0.796
Macro F1	0.792
Weighted F1	0.835
Strongest class	Neutral F1 = 0.988
Weakest class	Surprise F1 = 0.519

Critical limitation to state honestly

This is an image-level split. Similar frames or the same participant may occur on both sides, so 83.6 percent may overestimate performance on completely unseen people. A stronger experiment must group all images of one participant into only one of train, validation, or test.

8. Security algorithms and where they are used

Algorithm	Used at	Purpose
bcrypt	Signup, login, password change	One-way password hashing with salt.
JWT HS256	Authenticated API requests	Short-lived access identity and role claims.
SHA-256 token hash	Refresh-session table	Stores a non-reversible refresh-token fingerprint.
AES-256-GCM	After prediction, before storage	Encrypts image and prediction; GCM also detects ciphertext changes.
RSA-3072 OAEP-SHA256	Per scan	Wraps the random AES key so the AES key is not stored in plaintext.
SHA-256	Record and block creation	Produces fixed commitments and block hashes.
RSA-PSS-SHA256	Certificate export	Signs canonical certificate JSON.
Canonical JSON	Hashing and signing	Ensures the same logical fields produce the same bytes.

Envelope encryption

A new random 32-byte AES key is created for each scan. AES-GCM encrypts the raw image and prediction using separate random 12-byte nonces. The scan UUID is Additional Authenticated Data, binding ciphertext to its record. RSA-OAEP wraps the AES key. This design combines fast symmetric encryption with protected key distribution.

Why blockchain does not perform encryption

The blockchain stores hashes and links. It proves that committed bytes changed, but it does not hide the image. Confidentiality comes from AES-GCM; key protection comes from RSA-OAEP.

9. Private blockchain algorithm

Each block contains: **index, timestamp, record commitment, previous block hash, nonce, and block hash**. The record commitment covers encrypted image bytes, encrypted prediction bytes, and canonical metadata: scan ID, owner ID, timestamp, and source.

Block creation

- Read existing blocks in index order.
- Validate the complete chain before accepting a new record.
- Compute the record commitment with SHA-256.
- Set previous_hash to the latest block hash, or 64 zeroes for the first block.
- Increment nonce until the block hash starts with 00.
- Save upload, prediction, block, and audit event in one transaction.

Chain verification

- Index must match its sequence position.
- Every previous_hash must match the previous block.
- Recomputed block hash must equal the stored hash.
- Hash must satisfy the demonstration 00 prefix.
- Recomputed record commitment must equal block data_hash.
- AES-GCM must authenticate both stored ciphertexts.

Correct blockchain claim

The implemented chain is a simulated, centralized private blockchain that provides tamper evidence. It is not distributed consensus and is not unconditionally immutable. Stronger protection would require independent validator nodes or anchoring block hashes to an external ledger.

10. Signed PDF and JSON verification certificate

Certificate generation

- The authorized owner selects Certificate PDF or Signed JSON on a result page.

- The server includes scan ID, timestamp, source, prediction, all probabilities, model version, image hash, model hash, record commitment, and block proof.

- The server canonicalizes the JSON payload and computes SHA-256.

- The RSA private key signs the canonical bytes using RSA-PSS-SHA256.

- The PDF presents the same evidence and adds a QR link to the verification page.

Independent verification

Check	How it is verified
Digital signature	Use the installation public key to verify RSA-PSS over canonical JSON.
Report content	Recompute SHA-256 and compare with signed_payload_sha256.
Blockchain record	Compare scan commitment and block fields with the stored chain, then validate every link.
Model version	Hash the active model artifact and compare its SHA-256 with the certificate.

Tampering result

If a user changes even one probability, emotion label, hash, or block reference in the JSON, the canonical hash changes and RSA signature verification fails. The page displays which check failed rather than merely returning a general error.

Presentation sentence

The certificate makes the result portable and independently checkable: the PDF is readable by people, the JSON is machine-readable, the signature proves issuance, and the blockchain comparison proves the referenced stored record still matches.

11. Complete technology stack

Area	Technology	Why used
Frontend	React 19	Component-based interactive user interface.
Routing	React Router	Public, authenticated, result, administration, and verifier routes.
Build	Vite	Fast development and optimized portable build.
UI/visuals	CSS, Lucide React, Framer Motion, Recharts	Responsive design, icons, animation, probability and metric charts.
Backend	Python, FastAPI, Uvicorn	Typed high-performance API, validation, async upload and camera endpoints.
ORM/database	SQLAlchemy; SQLite local; PostgreSQL Docker	Portable development with a production-oriented database option.
Image processing	Pillow, OpenCV, NumPy	Image validation, preprocessing, HOG, camera frames.
Machine learning	scikit-learn; PyTorch baseline	Active logistic classifier and retained CNN comparison.
Security	cryptography, bcrypt, PyJWT	Authenticated encryption, RSA operations, password hashing, sessions.
Documents	ReportLab	PDF certificate, tables, QR code, pagination.
Camera	Browser MediaDevices, OpenCV, pyserial, WebSocket	UVC and server-side hardware capture options.
Testing	pytest, FastAPI TestClient	End-to-end API, security, tamper, camera, and certificate tests.
Deployment	Docker, Nginx, PostgreSQL option	Repeatable multi-service demonstration.

12. Database entities and important API stages

Entity	Stores
User	Name, email, bcrypt password hash, role, status, preferences.
Session	Refresh-token hash, expiry, revocation state, owning user.
Upload	Owner, source, timestamp, MIME, wrapped AES key, encrypted image.
Prediction	Encrypted result and associated block index.
Block	Commitment, previous hash, nonce, block hash.
Audit	Actor, action, time, IP, detail.
CameraTicket	Short-lived one-use authorization for WebSocket capture.

Representative API sequence

Route	Function
POST /api/preprocess	Validate image and return normalized preview.
POST /api/predictions	Run prediction, encryption, chain append, and atomic storage.
GET /api/predictions/{id}	Decrypt and return an authorized result.
POST /api/predictions/{id}/verify	Authenticate ciphertext and recompute record and chain proof.
GET /api/predictions/{id}/certificate.pdf	Create signed downloadable PDF.
GET /api/predictions/{id}/certificate.json	Create signed machine-readable certificate.
POST /api/certificates/verify	Verify uploaded JSON without requiring login.
GET /api/certificates/scan/{id}	Perform QR-linked live record verification.

13. How to demonstrate the project to the teacher

Recommended sequence: approximately 8 to 10 minutes.

Time	Show	What to say
0:00-1:00	Landing and About	State the problem, five labels, and Acquire -> Recognize -> Protect -> Verify.
1:00-2:30	Signup/login and Dashboard	Explain bcrypt, JWT, refresh rotation, roles, and record ownership.
2:30-4:00	Upload page	Choose a thermal image, show original and processed preview, then analyze.
4:00-5:00	Result page	Explain winning class, full probability distribution, and model limitation.
5:00-6:00	Verify integrity	Show SHA-256 commitment, block reference, AES-GCM, RSA-OAEP, and validation result.
6:00-7:00	Certificate export	Download PDF and JSON; scan QR or upload JSON on Verify Certificate page.
7:00-8:00	Blockchain explorer	Show blocks, previous hashes, and safe in-memory tamper demonstration.
8:00-9:00	Admin metrics	Show 83.6 percent test accuracy, macro F1, confusion matrix, class imbalance.
9:00-10:00	Limitations/future scope	Explain subject-disjoint evaluation, live-camera calibration, uncertainty, and multi-frame inference.

Avoid during the demo

Do not claim that blockchain encrypts data, that the model reads a person's true emotion, that 83.6 percent is guaranteed on unseen people, or that the local private chain is fully decentralized.

14. Common viva questions and strong answers

Question	Answer
Why thermal instead of RGB?	Thermal input represents infrared facial intensity and can operate without ordinary visible color. It also creates camera calibration and privacy challenges addressed by our pipeline.
Why HOG?	The supplied dataset is small. HOG provides stable local shape features and outperformed the initial small CNN on the measured split.
Why logistic regression?	It is fast, interpretable at the class-score level, provides multi-class probabilities, and performed well on the extracted HOG features.
Why not claim 83.6 percent for live use?	The evaluation is image-level and subject overlap is unknown. New people and cameras can cause domain shift.
What is CLAHE?	A local contrast enhancement algorithm with clipping that improves regional facial detail while limiting noise amplification.
Why AES-GCM?	It provides both confidentiality and authentication, so ciphertext alteration is detected.
Why RSA-OAEP?	It securely wraps each random AES data key using asymmetric encryption.
Why RSA-PSS?	It is a modern probabilistic digital-signature padding scheme used to sign certificate payloads.
What is stored in blockchain?	A hash commitment and provenance links, not the plaintext thermal image.
Is it decentralized?	No. It is a centralized private demonstration chain. Independent validators or external anchoring are future work.
What if the database changes?	Record commitment or block-link verification fails unless an administrator with full control rewrites and re-mines the whole local chain.
What happens below reliable confidence?	The current UI shows probabilities; recommended extension is an uncertainty threshold and multi-frame agreement before displaying a live class.
Can anyone download another user's image?	No. Server-side ownership and role checks protect record, image, and certificate exports.
What does the QR code verify?	It opens a public verifier for the referenced scan; uploaded JSON additionally verifies the exact signed certificate contents.

15. Honest limitations and meaningful future scope

Current limitations

- Image-level evaluation can contain participant overlap.
- The dataset is imbalanced, especially for surprise.
- New cameras may produce different palettes, temperature scales, resolutions, and noise.
- Per-image min-max normalization loses absolute thermal temperature.
- The active classifier does not detect whether an image is outside the training distribution.
- A single frame can be unstable; expressions and thermal measurements vary over time.
- The private chain is controlled by one installation.

Recommended next work

Priority	Improvement	Research value
1	Collect all five labels from each participant using the final thermal camera.	Reduces identity and camera confounding.
2	Use subject-disjoint train/validation/test splits.	Measures generalization to unseen people.
3	Add face detection, crop, alignment, and quality rejection.	Standardizes the model input.
4	Preserve calibrated radiometric temperature when hardware supports it.	Uses genuine thermal information.
5	Add out-of-distribution and confidence rejection.	Avoids forced unreliable labels.
6	Average 15-30 live frames and require class agreement.	Improves temporal stability.
7	Archive every deployed model by hash.	Allows historical certificate verification after model upgrades.
8	Anchor periodic chain hashes externally.	Strengthens integrity against a database administrator.

16. Final one-minute conclusion

Suggested conclusion

EmoTherm SecureChain demonstrates more than an emotion classifier. It is an end-to-end research evidence system. A thermal image is validated and normalized, HOG extracts facial-gradient structure, logistic regression estimates one of five expression classes, and the result is encrypted with a fresh AES key protected by RSA. A SHA-256 commitment enters a private blockchain, while roles and audit logs control access. Finally, a signed PDF and JSON certificate allows independent verification of the report, blockchain record, and model version. Our measured image-level holdout accuracy is 83.6 percent, but we clearly state that unseen-person and live-camera validation require subject-disjoint data and camera-specific calibration.

Key contribution statement

A privacy-preserving, tamper-evident, and independently verifiable workflow for thermal facial-expression research.

Remember these numbers

Item	Value
Supported classes	5
Files discovered	838
Usable unique images	825
Held-out test images	165
Measured accuracy	83.6%
Macro F1	0.792
AES data-key size	256 bits
RSA key size	3072 bits
Certificate signature	RSA-PSS-SHA256
Automated tests	15 passing

End of briefing